

Pittsburgh 311 Municipal Service Resolution

Assignment 1 corpus division and continuity to the final LLM API project

AGREED DATASET

Use the **WPRDC 311 Data Archive** as one fixed dataset package, with these two linked CSV resources:

- **311 Data:** request type, assigned department, status, origin, location, creation time, and closure time.
- **311 Issue and Category Codebook:** issue, exact **category**, department, codebook version, request type ID, and definition.

Join key: `request_type_id`. **Assignment rule:** use only the exact category values listed below, not keyword matching. The archive is historical and stopped updating after February 4, 2025, which makes it a stable corpus snapshot for the assignment.

Business question. Can an LLM-assisted 311 intake and routing API reduce the time required to resolve Pittsburgh service requests? The mechanism is *first-time-right intake*: correct classification, complete details, a targeted clarification question, and routing to the correct department.

— One Group Topic, Four Independently Filterable Corpora

Lead	Subtopic	Exact category values	Routing problem studied
1	Streets and Mobility	Road/Street Issues; Traffic and Street Sign Issues; Street Light; Pedestrian/Bicycle Concerns; Parking	Distinguish similar infrastructure problems.
2	Waste and Neighborhood Cleanliness	Garbage and Litter Issues; Neighborhood Issues; Graffiti Issues; Weeds/Debris	Separate missed service from illegal disposal and other conditions.
3	Buildings, Construction, and Accessibility	Building Maintenance; Construction Issues; Permits; Accessibility	Clarify ambiguous property reports and required evidence.
4	Parks, Trees, Animals, and Public Facilities	Parks Issues; Tree Issues; Animal Issues; City Facilities and Infrastructure	Route issues when ownership depends on location or urgency.

— Shared Assignment 1 Deliverable

Each member independently filters one subtopic and constructs a corpus using the same schema:

- **Labels:** `request_type_id`, issue, exact category, and responsible department.
- **Knowledge:** plain-language definition, inclusion/exclusion rules, confusable issues, required information, clarification questions, and official source.
- **Examples:** realistic resident complaints, including clear, ambiguous, incomplete, typo-heavy, and informal variants.
- **Operational evidence:** request volume plus median, 75th-percentile, and 90th-percentile resolution time for closed requests.

Source package: **WPRDC 311 Data Archive**. The dataset page identifies the service-request CSV and companion issue/category codebook as resources in the same archive.

Reproducible Corpus Split

The same dictionary should be shared by all four members to prevent taxonomy drift.

— Exact Python Category Filters

```
CATEGORY_GROUPS = {
    "streets_mobility": [
        "Road/Street Issues",
        "Traffic and Street Sign Issues",
        "Street Light",
        "Pedestrian/Bicycle Concerns",
        "Parking",
    ],
    "waste_neighborhood": [
        "Garbage and Litter Issues",
        "Neighborhood Issues",
        "Graffiti Issues",
        "Weeds/Debris",
    ],
    "buildings_construction": [
        "Building Maintenance",
        "Construction Issues",
        "Permits",
        "Accessibility",
    ],
    "parks_public_spaces": [
        "Parks Issues",
        "Tree Issues",
        "Animal Issues",
        "City Facilities and Infrastructure",
    ],
}

student_1_df = codebook[
    codebook["category"].isin(CATEGORY_GROUPS["streets_mobility"])
].copy()
```

— Minimum Cleaning and Join Contract

```
# Read IDs as strings so joins are not broken by numeric coercion.
requests = pd.read_csv("311_data.csv", dtype={"request_type_id": "string"})
codebook = pd.read_csv("311_issue_category_codebook.csv",
    dtype={"request_type_id": "string"})

codebook_clean = (
    codebook.loc[codebook["request_type_id"].notna()
        & codebook["category"].fillna("").ne("")]
    .sort_values("codebook_version")
    .drop_duplicates("request_type_id", keep="last")
)

requests_enriched = requests.merge(
    codebook_clean[["request_type_id", "issue", "category", "department"]],
    on="request_type_id", how="left", validate="many_to_one"
)
```

— Continuity to Final Project

The four corpora are concatenated into one retrieval knowledge base. The final API transforms:

Resident complaint → retrieve domain knowledge
→ classify + clarify + structure → routed 311 ticket

Suggested output: domain, issue, department, missing information, one clarification question, confidence, abstention flag, and historical resolution range.

— Evaluation Boundary

- **Primary business KPI:** median resolution time.
- **Proximal metrics:** issue and department accuracy, completeness, clarification rate, and abstention quality.
- Historical data can test routing quality and establish time-to-close baselines, but it cannot prove that the API causes faster resolution.
- A future staff-confirmed pilot is required for causal evidence.

Data discipline: exclude blank categories from the four-way split and report their count as unmapped coverage. Never place addresses or exact coordinates in the LLM corpus unless specifically required and privacy-reviewed.